

Parallel-support rank and a one-sided Merino–Welsh inequality

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Abstract

Let M be a finite matroid without loops or coloops, and let $P(M)$ be the union of its nontrivial parallel classes. We prove that

$$r_M(P(M)) \geq r(M) - 1 \implies T_M(0, 2) \geq T_M(1, 1).$$

The inequality is strict when $r_M(P(M)) = r(M) - 1$, and equality in the whole stated class occurs exactly for direct sums of copies of $U_{1,2}$. The dual statement holds for nontrivial series classes. In the graphic case, the hypothesis says that the vertex-spanning subgraph formed by parallel-edge bundles has at most two connected components. The proof combines an exact basis-activity formula for parallel extensions with deletion–contraction and an exponential surplus matching in the terminal two-element cocircuit core. The rank threshold is sharp if no further hypothesis is imposed: a binary rank-three matroid with parallel-support rank one violates the inequality. We also record a connected five-element counterexample at the same threshold, showing that connectivity alone does not extend the result to corank two.

1 Introduction

For a finite matroid M , the evaluations

$$T_M(2, 0), \quad T_M(0, 2), \quad T_M(1, 1)$$

are central quantities in the Merino–Welsh family of inequalities. The last evaluation is the number of bases. In the graphic case, the three evaluations count acyclic orientations, totally cyclic orientations, and spanning trees, respectively. The original graph conjecture asks whether a loopless, 2-connected graph satisfies [1]

$$\max\{T_M(2, 0), T_M(0, 2)\} \geq T_M(1, 1).$$

The corresponding assertion for all loopless, coloopless matroids is false, as are its additive and multiplicative variants [4]. Important positive results are nevertheless known for such classes as series–parallel graphs [2], split matroids [3], paving matroids [5], and matroids satisfying suitable density or cocircuit hypotheses [6]. The graph problem remains open.

This paper studies a one-sided inequality controlled by parallel structure. Write $P(M)$ for the union of all parallel classes of size at least two. If $P(M)$ spans M , every rank direction has a parallel substitute, and a direct activity comparison proves $T_M(0, 2) \geq T_M(1, 1)$. The main point is that one complete rank direction may be left outside that support. The remaining singleton elements need not be parallel to one another, and their number is unrestricted.

Our main theorem is the following.

Theorem 1.1. *Let M be a finite matroid without loops or coloops. If*

$$r_M(P(M)) \geq r(M) - 1,$$

then

$$T_M(0, 2) \geq T_M(1, 1). \tag{1.1}$$

If $r_M(P(M)) = r(M) - 1$, then (1.1) is strict. Equality in the full class of Theorem 1.1 occurs exactly when

$$M \cong \bigoplus_{i=1}^{r(M)} U_{1,2}, \tag{1.2}$$

with the empty direct sum allowed in rank zero.

By duality, the analogous statement holds when the nontrivial series classes span all but at most one rank direction in M^* , with $T_M(2, 0)$ in place of $T_M(0, 2)$.

The proof has two parts. First, we calculate the total contribution to $T_M(0, 2)$ of all bases projecting to a fixed basis of the simplification. Second, we induct on singleton parallel classes. At the final hyperplane core, bases with zero activity contribution are grouped with bases having several externally active parallel classes; the elementary inequality

$$2^K \geq 1 + K$$

pays for the entire group.

The threshold is exact for a condition depending only on $r_M(P(M))$. Section 5 gives a rank-three binary counterexample with $r_M(P(M)) = r(M) - 2$, and also a connected counterexample of order five. The latter shows that connectivity is not the missing hypothesis. A separate bounded analysis reduces a possible cosimple extension to three rank-two four-element quotient cores, but does not settle them; no corank-two or two-disjoint-bases theorem is claimed here.

2 Parallel classes and activities

Let $N = \text{si}(M)$ be the simplification of a loopless matroid M . For each $e \in E(N)$, let $k_e \geq 1$ be the size of the parallel class of M represented by e . Fix an order on $E(N)$, order the parallel classes accordingly, and within the class represented by e write

$$e_1 < e_2 < \cdots < e_{k_e}.$$

For a basis B of N , write $I(B)$ and $E(B)$ for its internally and externally active elements with respect to the chosen order.

Lemma 2.1. *The bases of M projecting to B contribute*

$$C_B = 2^{\sum_{f \in E(B)} k_f} \prod_{e \in I(B)} (2^{k_e} - 2) \prod_{e \in B \setminus I(B)} (2^{k_e} - 1) \quad (2.1)$$

to $T_M(0, 2)$. *The number of such bases is*

$$W_B = \prod_{e \in B} k_e. \quad (2.2)$$

Proof. A lifted basis is obtained by choosing one copy e_{j_e} from each class represented in B . Basic circuits and cocircuits behave uniformly inside a parallel class:

- if $e \in I(B)$, then the chosen copy is internally active exactly when $j_e = 1$;
- if $e \notin I(B)$, no chosen copy is internally active;
- the $j_e - 1$ smaller unchosen copies in a selected class are externally active;
- if $f \in E(B)$, every copy in the unselected class represented by f is externally active.

At $(x, y) = (0, 2)$, a basis with an internally active element contributes zero, while a basis with j externally active elements contributes 2^j . Summing independently over the copy chosen in each selected class gives

$$\sum_{j=2}^{k_e} 2^{j-1} = 2^{k_e} - 2$$

for an internally active class and

$$\sum_{j=1}^{k_e} 2^{j-1} = 2^{k_e} - 1$$

for a non-internally-active class. The unselected externally active classes give the leading power of two in (2.1). Formula (2.2) is immediate. $\square$

Proposition 2.2. *If every parallel class of a loopless matroid M has size at least two, then*

$$T_M(0, 2) \geq T_M(1, 1), \quad (2.3)$$

with equality exactly for direct sums of copies of $U_{1,2}$.

Proof. For every integer $k \geq 2$,

$$2^k - 2 \geq k, \quad 2^k - 1 > k.$$

Lemma 2.1 therefore gives $C_B \geq W_B$ for every basis B of the simplification. Summing over B proves (2.3).

For equality, every selected class in every basis must be internally active, every class size must be two, and no basis may have an externally active element. An ordered matroid with more

than one basis has a basis that is not entirely internally active. Thus the simplification has a unique basis. Since the matroid is loopless, its simplification is free, and (1.2) follows. The converse is immediate from multiplicativity of the Tutte polynomial. $\square$

3 A spanning parallel support

We first allow singleton classes while assuming that the nontrivial classes still span.

Proposition 3.1. *Let M be a finite loopless matroid. If $P(M)$ spans M , then*

$$T_M(0, 2) \geq T_M(1, 1).$$

If M has a singleton parallel class, the inequality is strict.

Proof. Induct on the number of singleton parallel classes. Proposition 2.2 is the base case. Choose a singleton element e . Since the nontrivial parallel classes span, e is neither a loop nor a coloop. Deletion preserves the spanning parallel support. Contraction creates no loop because e has no parallel mate; the images of the old nontrivial classes still span, and some classes may merge. Both minors have fewer singleton classes, so deletion–contraction and induction prove the inequality.

It remains to justify strictness in the only potentially marginal step. If there is exactly one singleton e and $M \setminus e$ is an equality case, then $M \setminus e$ is a direct sum of binary parallel classes. A fundamental circuit of e meets at least two of those classes. If it meets exactly two, contracting e merges them into a parallel class of size at least four; if it meets at least three, the simplification of M/e has a circuit. In either case M/e is not an equality case of Proposition 2.2. The contraction term is therefore strict. Strong induction gives strictness whenever a singleton class is present. $\square$

4 Parallel support of corank one

We now prove Theorem 1.1. The spanning case is Proposition 3.1, so assume

$$r_M(P(M)) = r(M) - 1.$$

Put

$$F = \text{cl}_M(P(M)).$$

Then F is a hyperplane. Since M has no coloops, $E(M) \setminus F$ has at least two elements.

We again induct on the number of singleton parallel classes.

4.1 Removing singleton classes inside the hyperplane

Suppose first that $e \in F$ is a singleton. It is neither a loop nor a coloop. Since $e \in \text{cl}_M(P(M))$, deletion leaves the old nontrivial classes with rank $r(M) - 1$. No element of F becomes a coloop: the nontrivial classes still span F . Nor does an element outside F become a coloop, because at least two such elements remain and each extends the hyperplane.

After contracting e , the old nontrivial classes have rank $r(M) - 2$ in the rank- $r(M) - 1$ matroid M/e . Contraction creates no loop because e was a singleton, and contraction preserves the absence of coloops. Thus both minors satisfy the hypothesis of Theorem 1.1 and have fewer singleton classes. The deletion minor remains in the corank-one layer, so its inductive inequality is strict.

We may therefore assume that every element of F belongs to a nontrivial parallel class.

4.2 Removing surplus elements outside the hyperplane

Every element outside F is a singleton, since a nontrivial parallel class would be contained in $P(M) \subseteq F$. If there are at least three such elements, choose $e \notin F$. Deleting e leaves at least two elements outside F , so the deletion minor remains loopless, coloopless, and in the corank-one layer. In M/e , the image of F has full rank, so Proposition 3.1 applies. Induction and deletion-contraction prove the result, and the deletion term is strict.

It remains only to treat

$$E(M) \setminus F = \{p, q\}, \quad (4.1)$$

with every class in F nontrivial.

4.3 The terminal core

Order all classes in F before $p < q$. Let A be a basis of the simplification restricted to F . Both $A \cup \{p\}$ and $A \cup \{q\}$ are bases.

For $A \cup \{p\}$, the element p is internally active: the only possible replacement outside the hyperplane is q , which is later. Hence the whole lifted basis block contributes zero to $T_M(0, 2)$. For $A \cup \{q\}$, the earlier element p replaces q , and Lemma 2.1 shows that the block contribution is at least its basis weight. It remains to pay for the zero block associated with $A \cup \{p\}$.

Take the fundamental circuit of q with respect to $A \cup \{p\}$. It contains p , since $A \cup \{q\}$ is a basis. It also contains a class from F , because p and q cannot be parallel. Let $h(A)$ be the least such class and put

$$B(A) = A \setminus \{h(A)\} \cup \{p, q\}. \quad (4.2)$$

Basic exchange shows that $B(A)$ is a basis. Relative to this basis, $h(A)$ is externally active, and both p and q fail to be internally active because either may be replaced through $h(A)$.

Fix a basis B containing p, q , and let X_B be the set of classes $h(A)$ from all bases A mapped to B . Write

$$W_B = \prod_{e \in B \cap F} k_e, \quad K_B = \sum_{h \in X_B} k_h.$$

All classes in X_B are externally active at B . Lemma 2.1 therefore gives

$$C_B \geq 2^{K_B} W_B. \quad (4.3)$$

The basis B itself and all zero blocks mapped to it contain altogether

$$W_B + \sum_{h \in X_B} k_h W_B = (1 + K_B) W_B \quad (4.4)$$

bases. Since $2^{K_B} \geq 1 + K_B$, (4.3) pays for (4.4). If $X_B \neq \emptyset$, then $K_B \geq 2$, and the inequality is strict.

A basis containing p, q with no preimage is also paid for by its own activity block. Indeed, its F -part has size $r(M) - 2$ and can be completed to a basis of F by an earlier class; that class can replace either p or q , so neither singleton is internally active. Lemma 2.1 gives $C_B \geq W_B$.

All basis blocks have now been paid for, and at least one block of the form (4.2) occurs. The core inequality is strict. This completes the induction and the proof of Theorem 1.1. $\square$

5 Equality, duality, and sharpness

The corank-one case is always strict. In the spanning case, Proposition 3.1 and Proposition 2.2 show that equality occurs precisely for (1.2). This proves the equality statement in Theorem 1.1.

The identity

$$T_{M^*}(x, y) = T_M(y, x)$$

applied to Theorem 1.1 gives the dual theorem for nontrivial series classes.

The rank threshold cannot be lowered using only the rank of $P(M)$. Over $\text{GF}(2)$, consider the column matroid of

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

This matroid is $U_{2,3} \oplus U_{1,2}$. It is loopless and coloopless, has rank three, and its only nontrivial parallel class has rank one. Hence

$$r_M(P(M)) = 1 = r(M) - 2,$$

but

$$T_M(0, 2) = 4 < 6 = T_M(1, 1). \quad (5.1)$$

Connectivity alone does not repair (5.1). Duplicate one element of $U_{3,4}$; equivalently, take the binary column matroid

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

It is connected, loopless, and coloopless, with parallel-support rank one. Deletion–contraction along a member of the parallel pair gives

$$T_M(x, y) = x^3 + x^2 + x + y + x^2y + xy + y^2,$$

so

$$T_M(0, 2) = 6 < 7 = T_M(1, 1). \quad (5.2)$$

This second example is not cosimple: its three singleton elements form two-element cocircuits in pairs.

For context, suppose that $F = \text{cl}_M(P(M))$ has rank $r(M) - 2$ and that M is cosimple. The cocircuits of the rank-two quotient $Q = M/F$ are the cocircuits of M contained in $E(M) \setminus F$, so Q is cosimple. In a rank-two matroid, the complement of each parallel class is a cocircuit. A direct case split on the number of parallel classes therefore gives two disjoint bases of Q : choose from four classes if there are at least four; with three classes, cosimplicity supplies two classes with two available elements; with two classes, each class has at least three elements. A basis of F can be chosen inside $P(M)$, and replacing each member by a parallel mate gives a disjoint second basis of F . Combining these pairs lifts the two disjoint bases of Q to two disjoint bases of M .

This observation also gives a finite structural reduction. The induction is made in the larger class of loopless, coloopless matroids with corank-two parallel support for which Q contains two disjoint bases; cosimplicity is not imposed on the minors. Fix two such bases of Q . If $|E(Q)| > 4$, choose an element e outside their union. Their lifts give two disjoint bases of M avoiding e , so $M \setminus e$ retains full rank, has no coloop, and remains in the same corank-two support class. Also $e \notin F$, hence it is a singleton and M/e is loopless; contraction preserves colooplessness. The old parallel support has rank $r(M) - 2$ in M/e , one less than the new matroid rank, so Theorem 1.1 controls the contraction term. Induction and deletion–contraction therefore reduce a possible theorem for this layer to four-element rank-two quotients. Their relevant parallel types are $U_{2,4}$, $(2, 1, 1)$, and $(2, 2)$. The required quantitative activity matching in those cores remains open in the present work. Thus (5.2) is used only to show the sharpness of the rank-only and connectivity extensions; no cosimple corank-two or two-disjoint-bases result is asserted.

6 Graphic interpretation

Let H be a connected loopless multigraph. Form a simple graph $D(H)$ on the same vertex set by retaining an edge uv exactly when H has at least two parallel edges between u and v . The rank of the union of nontrivial parallel classes in the cycle matroid is

$$|V(H)| - c(D(H)),$$

where $c(D(H))$ is the number of connected components, including isolated vertices. Since $r(M(H)) = |V(H)| - 1$, Theorem 1.1 gives the following.

Corollary 6.1. *If H is connected and bridgeless and $D(H)$ has at most two connected components, then the number of totally cyclic orientations of H is at least its number of spanning trees. If $D(H)$ has exactly two components, the inequality is strict.*

In particular, one may start with an arbitrary connected simple graph, select a spanning forest with at most two components, and duplicate every edge of that forest. All remaining edges may retain multiplicity one or be thickened arbitrarily. The resulting bridgeless multigraph satisfies the one-sided Merino–Welsh inequality whenever the stated no-bridge condition holds. The connected counterexample (5.2) is graphic: it is a four-cycle with one edge duplicated. Its heavy-edge support has three components, so the two-component threshold in Corollary 6.1 is also sharp under connectivity alone.

7 Computational checks

The proof is symbolic. The accompanying exact Python verifier checks the parallel-class activity formula, the corank-one induction interfaces, equality controls, and both corank-two counterexamples on finite low-rank instances. It uses explicit failures rather than Python assertions, so optimization mode does not remove the checks.

From the published verification directory, run:

```
python -B verify_corank_one_parallel_support.py
```

```
python -B -O verify_corank_one_parallel_support.py
```

These computations test formulas and boundary examples; they are not a proof of Theorem 1.1.

8 Scope and prior work

Noble and Royle proved the multiplicative Merino–Welsh inequality for series–parallel graphs [2]. Ferroni and Schröter proved it for split matroids [3]. The class in Theorem 1.1 is not contained in either class: for example, doubling every edge of K_5 satisfies the spanning-support hypothesis, is not series–parallel, and has $M(K_5)$ as a minor; since split matroids are minor closed and $M(K_5)$ is not split, this example is not split.

Kung’s density and cocircuit conditions [6] cover some dense or large-cocircuit subfamilies but not the full sparse class considered here. Csikvári recently improved a related Tutte-evaluation inequality used in this area [7]. The activity and deletion–contraction methods are standard; the proposed contribution is the parallel-support rank threshold, its strict corank-one form, the complete equality classification, and the sharp rank-two-support boundary.

The theorem proves only the one-sided inequality $T_M(0, 2) \geq T_M(1, 1)$ on the stated class. It does not prove the additive or multiplicative strengthenings. In fact, the uniformly doubled matroids used by Beke, Csáji, Csikvári and Pituk [4] belong to the spanning-support layer and show that the multiplicative inequality cannot be added to Theorem 1.1 in general. Nor does this paper prove the full graph conjecture or the general conjecture for matroids with two disjoint bases.

Directed searches completed on 4 September 2026 found no source stating the same parallel-support rank theorem. This is a bounded novelty assessment, not a guarantee of global priority or a substitute for external peer review.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

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